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CAPSTONE PROJECT

NOTEBOOK 22: FINAL RESULTS SUMMARY AND COMPARISON

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Purpose:

This notebook consolidates the results from the structured,

audio, hybrid, and inference stages of the capstone project.

The goal is to:

1. Summarize the 8-genre validated system
2. Summarize the expanded 8-genre experiment
3. Summarize the 10-genre broader extension
4. Compare final structured, audio, and hybrid results

5. Export clean summary tables and plots for reporting

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In [1]:

```
# =====
# 1. IMPORT LIBRARIES
# =====

import os
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:

```
# =====
# 2. CREATE FINAL RESULTS TABLE
# =====

final_results = pd.DataFrame([
    {
        "Scenario": "8-Genre Validated System",
        "Structured Model": "Random Forest",
        "Structured Accuracy": 0.4053333333333333,
        "Audio Model": "Improved Audio CNN",
        "Audio Accuracy": 0.4533333480358124,
        "Hybrid Accuracy": 0.4613333333333333,
        "Notes": "Main validated proof-of-concept pipeline"
    },
    {
        "Scenario": "Expanded 8-Genre Experiment",
        "Structured Model": "Random Forest",
        "Structured Accuracy": 0.4133333333333333,
        "Audio Model": "Improved Audio CNN",
        "Audio Accuracy": 0.41333332657814026,
        "Hybrid Accuracy": 0.44266666666666665,
        "Notes": "More training data alone did not improve final hybrid performance"
    },
    {
        "Scenario": "10-Genre Broader Extension",
        "Structured Model": "SVM",
        "Structured Accuracy": 0.43177966101694915,
        "Audio Model": "Improved Audio CNN",
        "Audio Accuracy": 0.31409168243408203,
        "Hybrid Accuracy": 0.47156196943972833,
        "Notes": "Broader label space with full validation/test coverage"
    }
])
```

```

])

print("Final Results Summary:")
display(final_results)

```

Final Results Summary:

	Scenario	Structured Model	Structured Accuracy	Audio Model	Audio Accuracy	Hybrid Accuracy	Notes
0	8-Genre Validated System	Random Forest	0.405333	Improved Audio CNN	0.453333	0.461333	Main validated proof-of-concept pipeline
1	Expanded 8-Genre Experiment	Random Forest	0.413333	Improved Audio CNN	0.413333	0.442667	More training data alone did not improve final...
2	10-Genre Broader Extension	SVM	0.431780	Improved Audio CNN	0.314092	0.471562	Broader label space with full validation/test ...

In [3]:

```

# =====
# 3. SAVE FINAL RESULTS TABLE
# =====

os.makedirs("../data/processed", exist_ok=True)

final_results.to_csv(
    "../data/processed/final_results_summary.csv",
    index=False
)

print("Saved final results summary to ../data/processed/final_results_summary.csv")

```

Saved final results summary to ../data/processed/final_results_summary.csv

In [4]:

```

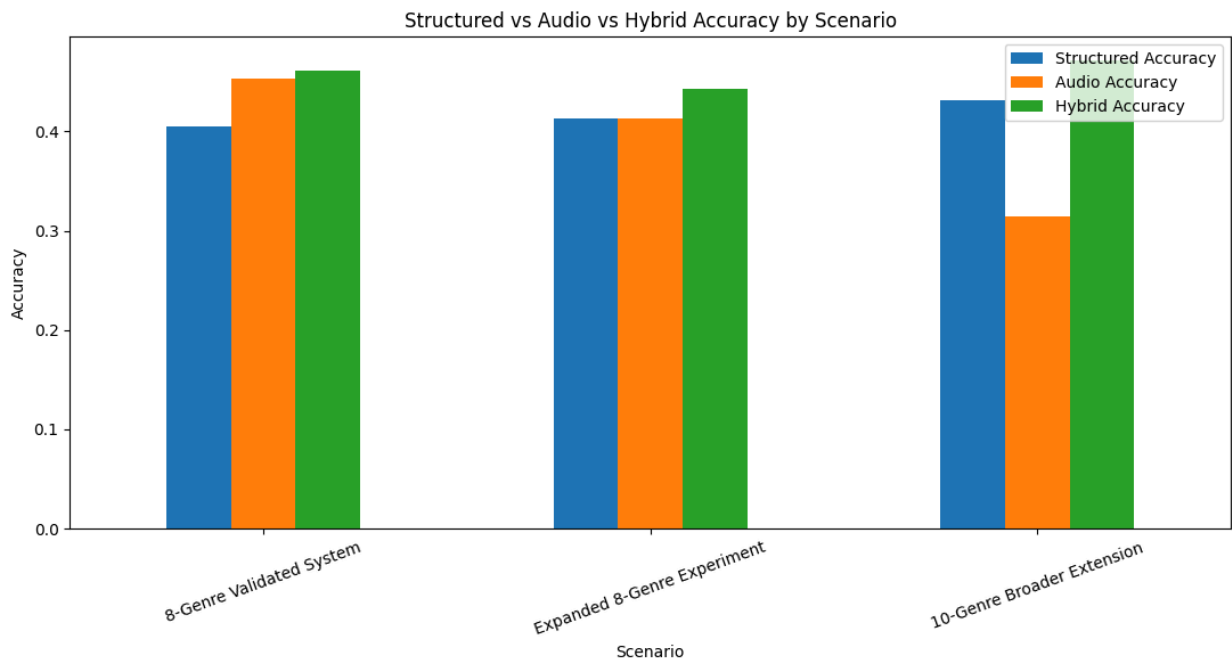
# =====
# 4. PLOT STRUCTURED VS AUDIO VS HYBRID BY SCENARIO
# =====

plot_df = final_results.set_index("Scenario")[
    ["Structured Accuracy", "Audio Accuracy", "Hybrid Accuracy"]
]

ax = plot_df.plot(kind="bar", figsize=(11, 6))
ax.set_ylabel("Accuracy")
ax.set_title("Structured vs Audio vs Hybrid Accuracy by Scenario")
plt.xticks(rotation=20)

```

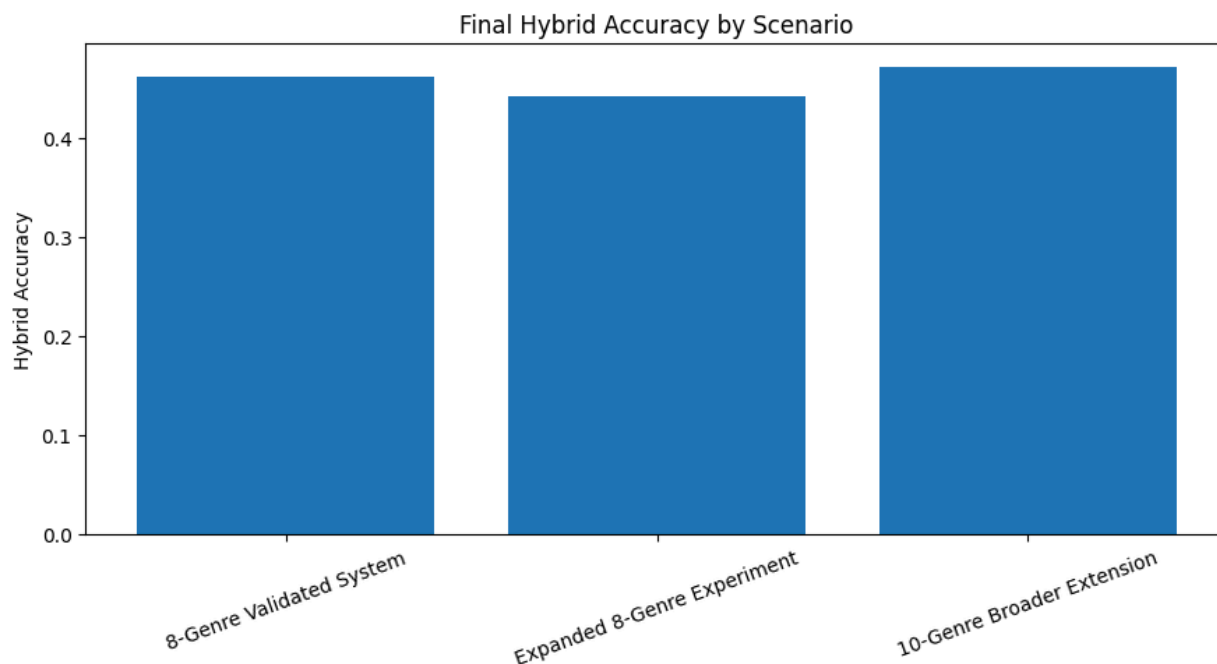
```
plt.tight_layout()
plt.show()
```



In [5]:

```
# =====
# 5. PLOT HYBRID ACCURACY ONLY
# =====

plt.figure(figsize=(9, 5))
plt.bar(final_results["Scenario"], final_results["Hybrid Accuracy"])
plt.ylabel("Hybrid Accuracy")
plt.title("Final Hybrid Accuracy by Scenario")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



In [6]:

```
# =====
# 6. CREATE FINAL INTERPRETATION TABLE
# =====

interpretation_df = pd.DataFrame([
    {
        "Finding": "Hybrid fusion was consistently the best approach",
        "Meaning": "Across both the 8-genre and 10-genre systems, combining
structured and audio branches gave the strongest final performance."
    },
    {
        "Finding": "The expanded 8-genre experiment did not improve the final
model",
        "Meaning": "Increasing training data alone was not enough to improve the
hybrid pipeline under the current setup."
    },
    {
        "Finding": "The 10-genre extension remained viable",
        "Meaning": "Even on the broader and harder 10-genre task, the hybrid model
still outperformed the individual structured and audio branches."
    },
    {
        "Finding": "The deployed pipelines worked end to end",
        "Meaning": "Final inference notebooks successfully accepted audio input,
extracted features, generated branch predictions, fused them, and returned final
genre outputs."
    }
])

print("Final Interpretation Table:")
display(interpretation_df)
```

Final Interpretation Table:

	Finding	Meaning
0	Hybrid fusion was consistently the best approach	Across both the 8-genre and 10-genre systems, ...
1	The expanded 8-genre experiment did not improv...	Increasing training data alone was not enough ...
2	The 10-genre extension remained viable	Even on the broader and harder 10-genre task, ...
3	The deployed pipelines worked end to end	Final inference notebooks successfully accepte...

In [7]:

```
# =====
# 7. SAVE FINAL INTERPRETATION TABLE
# =====

interpretation_df.to_csv(
    "../data/processed/final_interpretation_summary.csv",
    index=False
)

print("Saved interpretation summary to
../data/processed/final_interpretation_summary.csv")
```

Saved interpretation summary to ../data/processed/final_interpretation_summary.csv

In [8]:

```
# =====
# 8. OPTIONAL: INCLUDE BATCH DEMO SUMMARY
# =====

batch_demo_summary = pd.DataFrame([
    {
        "Demo Type": "10-Genre Batch Inference Demo",
        "Purpose": "Qualitative deployment demonstration",
        "Demo Top-1 Accuracy": 0.1333333333333333,
        "Interpretation": "Useful for showing pipeline behavior on multiple files,
but not the primary model evaluation metric."
    }
])

print("Batch Demo Summary:")
display(batch_demo_summary)

batch_demo_summary.to_csv(
    "../data/processed/batch_demo_summary.csv",
    index=False
)
```

```
print("Saved batch demo summary to ../data/processed/batch_demo_summary.csv")
```

Batch Demo Summary:

	Demo Type	Purpose	Demo Top-1 Accuracy	Interpretation
0	10-Genre Batch Inference Demo	Qualitative deployment demonstration	0.133333	Useful for showing pipeline behavior on multip...

Saved batch demo summary to ../data/processed/batch_demo_summary.csv

```
In [9]: # =====
# 9. OPTIONAL: LOAD SAVED BATCH DEMO GENRE SUMMARY IF PRESENT
# =====

batch_demo_summary_path =
"../data/processed/batch_inference_demo_10genre_summary.csv"

if os.path.exists(batch_demo_summary_path):
    batch_demo_by_genre = pd.read_csv(batch_demo_summary_path)
    print("Loaded batch inference demo genre summary:")
    display(batch_demo_by_genre)
else:
    print("No saved batch_inference_demo_10genre_summary.csv found in
../data/processed")
```

Loaded batch inference demo genre summary:

	actual_genre	samples	correct_predictions	accuracy
0	Classical	3	0	0.000000
1	Electronic	3	0	0.000000
2	Experimental	3	1	0.333333
3	Folk	3	0	0.000000
4	Hip-Hop	3	0	0.000000
5	Instrumental	3	0	0.000000
6	International	3	0	0.000000
7	Pop	3	0	0.000000
8	Rock	3	0	0.000000
9	Spoken	3	3	1.000000